Data Engineering Technical Assignment- Expernetic

Inside Airbnb · Cape Town · Scrape 28–29 September 2025

Submitted By:

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1. Executive Summary

Cape Town is not a peer-to-peer "sharing economy." It is a mature, professionalised, coastal-premium short-term rental market, and the data says so unambiguously. This report analyses a single-city snapshot of the public Inside Airbnb dataset (scrape dated 28–29 September 2025: 26,877 listings, 9.8 million calendar-day records, and 664,377 reviews) and converts it into decision-ready intelligence for pricing, supply strategy, and platform operations. All monetary figures are in South African Rand (ZAR).

The market is structurally defined by asset class and geography, not host effort. Entire-home listings command a mean nightly rate of R2,938 against R1,408 for private rooms — a statistically decisive gap (Welch's t, $p \approx 0$) with a medium real-world effect size (Cohen's $d = 0.54$). Location is the second structural lever: a listing's Ward explains roughly 12.9% of all price variance (ANOVA $\eta^2 = 0.129$), and the premium is coastal rather than central — the correlation between price and distance to the central business district is effectively zero ($r = -0.076$). Cape Town prices radiate from the Atlantic Seaboard, not from downtown. By contrast, the levers hosts obsess over are marginal: the Superhost badge lifts review scores by only 0.19 points ($d = 0.47$), and accumulating reviews actually coincides with lower prices ($d = -0.31$), because review volume is a by-product of high-turnover budget inventory, not pricing power.

Supply is dangerously concentrated. The host base carries a Gini coefficient of 0.430; the top 10% of hosts control 43.7% of all listings, and operators running six or more properties hold 33.1% of city inventory. This is simultaneously a churn risk (losing ~1,100 top hosts removes a large share of supply) and a regulatory exposure (a municipal cap on commercial operators would erase a third of inventory overnight).

Demand is large, growing, and violently seasonal. Using reviews as a booking proxy, post-COVID monthly demand runs roughly 7× (744%) above pre-pandemic levels. Within the year, January generates 124.5% more activity than the June trough, and forward calendar availability tightens from ~65% in winter to ~35% across the December–January peak. A static, year-round pricing strategy leaves substantial revenue unclaimed.

Quality scores have lost their signalling power. The median review score is 4.89, with 89.2% of reviewed listings scoring 4.5 or higher — a 4.5 actually sits in the bottom ~11% of active inventory. Of the sub-dimensions, "Value" and "Accuracy" (not "Location") are the true drivers of overall satisfaction ($r = 0.87$ and 0.86). Guests reward listings that price fairly and match their photos.

What actually moves price is physical capacity. A multivariate regression ($R^2 = 0.515$) shows an additional bathroom is worth ~R1,299/night and each extra guest accommodated ~R393/night, while bedroom count becomes statistically irrelevant once capacity is controlled for. Capital invested in capacity returns far more than effort spent chasing badges or ratings.

Priority recommendations for the consultancy's stakeholders:

- **Revenue strategy:** deploy seasonal dynamic pricing as a default, not an option — a seasonal multiplier (roughly double for Q1, halve for the winter trough) is required for any credible revenue forecast.

- **Supply / account management:** white-glove the top ~1,100 hosts to defend against churn, and intervene on the ~6,720 "inactive" listings that appear overpriced relative to the ~R1,310 market-clearing rate to reactivate dead inventory.
- **Automated valuation:** weight capacity features and Ward heavily, engineer a distance-to-coastline feature (CBD distance is noise), and move to a log-transformed or tree-based model — the linear model's error variance fans out badly at the luxury end.
- **Product / trust:** rely on derived badges and sub-scores ("Value", "Accuracy") rather than the inflated headline rating, and concentrate quality audits on absentee-managed entire homes in the 4.1–4.3 "danger zone."

Data constraint. Calendar prices are 100% null in this scrape, so true daily, weekend-versus-weekday, and dynamic seasonal pricing cannot be measured directly. We report this honestly rather than substitute placeholder values, and flag it as a data-source risk for any production revenue tool built on Inside Airbnb data for Cape Town. The flagship Open Innovation deliverable (Section 10) is an interactive Power BI command center that makes these findings explorable for non-technical stakeholders.

2. Objectives & Scope

Objective. Acting as the data engineer and analyst for a hypothetical Airbnb market-intelligence consultancy, the goal was to transform a raw, public short-term-rental dataset into reproducible engineering artifacts, defensible analytical findings, and actionable business recommendations for product managers, revenue strategists, and operations leads. Concretely, that meant building a Bronze → Silver → Gold ("medallion") pipeline, modelling the result as an analytical warehouse, conducting rigorous exploratory and statistical analysis, and surfacing the insight through both an AI-assisted data copilot and a business-facing dashboard.

City selection: Cape Town, single market. The Inside Airbnb catalogue spans dozens of global cities, and the brief explicitly rewards both depth (one city, exceptional rigour) and breadth (multi-city scalability). We chose depth. A single, complex, highly seasonal market analysed thoroughly — with every assumption, limitation, and trade-off documented — demonstrates the engineering judgement and analytical storytelling the rubric weights most heavily, and avoids the shallow coverage the brief warns against. Cape Town is a particularly instructive single market: its Southern-Hemisphere seasonality, coastal pricing geography, and heavily commercialised host base expose dynamics a flatter market would hide.

Prioritisation rationale. Effort was deliberately concentrated on the sections where depth compounds into credibility:

- **Core, mandatory depth (Sections 02–05):** dataset familiarisation, data engineering, exploratory data analysis, and statistical analysis — the foundation every downstream claim rests on.
- **High-value extensions (Sections 07–08):** an LLM-powered data copilot for natural-language querying, and a Power BI market-intelligence dashboard (supply density and price heatmap).
- **Deliberately deferred (Section 06, Data Science / ML):** predictive modelling and clustering were not attempted within the one-week budget. This is a conscious prioritisation choice in line with the brief's own design philosophy, recorded transparently in the project decision log and in Section 13. The statistical regression in Section 7 already establishes the price-driver groundwork a future modelling effort would build on.

Scope boundaries. Because the analysis is single-city, the multi-city comparison methods named in the brief (Bonferroni/FDR corrections across markets, cross-market effect-size comparison, cross-city model transfer) are not applicable and are scoped out by design rather than omitted by oversight. The single most consequential scope boundary is dataset-imposed, not chosen: the null calendar-price field, which removes direct dynamic-pricing analysis from the achievable set and is treated as a first-class limitation throughout.

3. Dataset Overview

Source and snapshot. All data is sourced exclusively from Inside Airbnb, an independent public project that scrapes and republishes Airbnb listing data. This analysis uses the Cape Town scrape dated 28–29 September 2025. No external or supplementary data was introduced.

Files used and their grain. Five files form the raw (Bronze) layer:

File	Grain & contents
listings.csv	26,877 rows × 79 columns; one row per listing. The core entity — price, room/property type, host attributes, location, capacity, amenities, review-score summaries.
calendar.csv	9,810,109 rows × 7 columns; one row per listing per future date, over a forward-looking 365-day window (28 Sep 2025 → 28 Sep 2026).
reviews.csv	664,377 rows × 6 columns; one row per review, spanning 15 Jun 2010 → 28 Sep 2025.
neighbourhoods.csv	116 rows × 2 columns; neighbourhood reference list, where Cape Town neighbourhoods are encoded as electoral "Ward" identifiers.
neighbourhoods.geojson	Polygon boundaries for those Wards, used for spatial aggregation.

Entities and relationships. The schema is a clean star around the listing. listings.id is the primary key. calendar.listing_id and reviews.listing_id are foreign keys referencing it, and referential integrity is intact in both directions — zero orphaned calendar rows and zero orphaned reviews, with every listing represented in the calendar. Of the 26,877 listings, 20,753 (77%) carry at least one review and 6,124 (23%) have none — the latter being either brand-new "cold-start" inventory or dormant listings. In business terms, each listing is a bookable unit, each host (identified by host_id) may control many listings, each calendar row is a single future night's availability, and each review is one completed-stay signal used here as the best available proxy for realised demand.

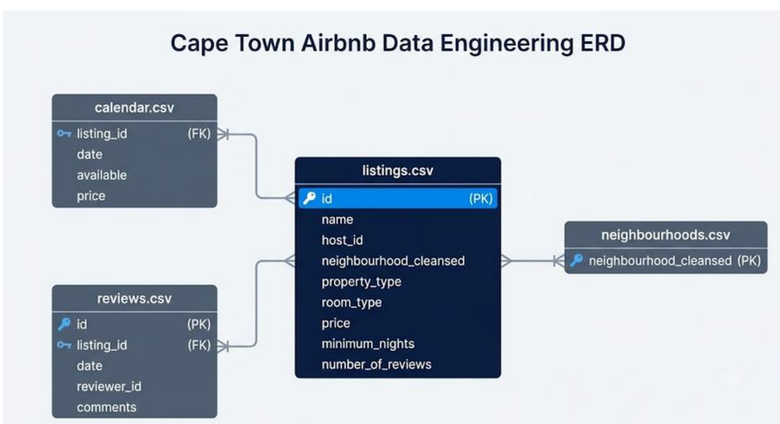

Figure 1 Source Dataset Entity-Relationship Diagram (ERD)

Limitations (data-level, not implementation gaps). Several constraints shape — and bound — every finding:

- **Calendar pricing is 100% null (dominant constraint).** Both price and adjusted_price are empty across all 9.8M calendar rows. Verified as a genuine source gap, not a parsing artifact; this removes direct daily, weekend/weekday, and dynamic seasonal price analysis from scope (Hypothesis H5 is reported as untestable).
- **Structurally empty fields.** neighbourhood_group_cleansed and calendar_updated are 100% null; license is 99.5% null and host_neighbourhood 99.3% null — none can support analysis.
- **Review-dependent missingness.** The review-score family and reviews_per_month are ~22.8% null, corresponding almost exactly to the 23% of listings with no reviews — missing not at random, but for a known and explainable reason.
- **Price missingness.** price is null for 16.37% of raw listings; these are excluded from price-based analysis, leaving 22,476 priced listings in the Gold layer.
- **Coordinate jitter.** Inside Airbnb deliberately offsets each listing's latitude/longitude by up to ~150 m for host privacy. Neighbourhood-level aggregation is reliable; street-level pinpointing is not.
- **Residual anomalies (carried to Section 13).** Isolated implausible values such as a bedrooms maximum of 50 and a review_scores_rating maximum of 19.88 (outside the valid 0–5 range) survive in a small number of rows and are noted as caveats on the affected fields.

Assumptions documented before analysis. To avoid silent, untraceable decisions:

- **Currency.** Prices are treated as ZAR; the "$" prefix in the raw strings is an Inside Airbnb formatting convention, not US dollars. Raw values also carry thousands separators (e.g., $2,315.00) requiring cleaning before numeric casting.
- **Bathrooms source.** Where numeric bathrooms is null (16.3%) but bathrooms_text is populated (only 0.9% null), the text field is the more reliable source — it recovers 4,150 otherwise-missing values.
- **Duplicates.** Zero exact-row and zero primary-key duplicates exist. A geospatial fuzzy check surfaced 11,956 candidate pairs (same host, <50 m apart) across 1,272 hosts; these are interpreted as legitimate multi-unit properties and are flagged in the Silver layer, not deleted.
- **Availability semantics.** A calendar day marked unavailable (available = 'f') conflates genuine guest bookings with manual host blocks; the two cannot be separated. All occupancy figures derived from it are therefore explicit upper bounds.
- **Seasonal pricing proxy.** Because calendar prices are null, the month of each listing's last_review is used as a directional proxy for when a static price was active — an approximation with a known upward bias, flagged wherever it appears.

4. Methodology

The analytical approach followed a single guiding principle: understand the data before transforming it, transform it reproducibly before analysing it, and report what the data can support rather than what the brief asks for. Methodologically this resolved into four deliberate choices.

A medallion (Bronze → Silver → Gold) data architecture. Rather than clean the data in a single ad-hoc script, the work was structured into three progressively refined layers. Bronze holds the raw source files exactly as scraped, never modified. Silver holds cleaned, validated, type-cast data. Gold holds the joined, enriched, business-ready tables that every notebook reads from. This choice was made for traceability and reproducibility: each stage can be re-run independently, every output column can be traced back to a source transformation, and the analytical notebooks are insulated from raw-data messiness. The engineering detail of each layer is documented in Section 5.

Profiling-first investigation. Before any cleaning logic was written, a dedicated profiling pass documented row counts, null rates, cardinality, referential integrity, duplicate structure, outliers, and domain-validity (no negative or zero prices, valid coordinates, minimum_nights ≤ maximum_nights). Every assumption about ambiguous fields — currency denomination, the bathrooms versus bathrooms_text reliability question, the fuzzy-duplicate interpretation — was settled and recorded against evidence at this stage, not improvised during analysis.

Assumption-driven statistical test selection. Because the price variable is severely right-skewed and review scores are ceiling-clustered, parametric defaults could not be applied blindly. For every hypothesis, normality was checked with Shapiro-Wilk and equal variance with Levene's test; both were formally rejected in every price- and score-based comparison. The response was to pair the variance-robust Welch's t-test with the distribution-free Mann-Whitney U test as a robustness check, and to report both. For the multi-group neighbourhood comparison, one-way ANOVA with a Tukey HSD post-hoc correction was used. Critically, effect sizes (Cohen's d, eta-squared) were reported alongside every p-value, because with 22,000+ listings even trivial differences return significant p-values — effect size is what separates a result that matters commercially from one that is merely detectable.

Multicollinearity control in the price model. The correlation analysis revealed that the strongest price predictors — accommodates ($r = 0.64$), bathrooms_final ($r = 0.63$), and bedrooms ($r = 0.58$) — are also heavily correlated with each other (accommodates↔bedrooms $r = 0.86$; bedrooms↔bathrooms $r = 0.83$). A naïve correlation reading would therefore triple-count the same underlying "size" effect. To isolate each feature's true marginal contribution, a multivariate OLS regression was fitted, and Variance Inflation Factors (VIF) were computed to quantify the redundancy directly. The VIF diagnostics confirmed the problem (accommodates VIF = 21.6, bedrooms = 17.6, bathrooms_final = 12.3 — all well above the severe threshold of 10), and the regression itself resolved it: once capacity was controlled for, bedrooms collapsed to statistical insignificance ($p = 0.228$), confirming that bedroom count carries no independent pricing information beyond what accommodates and bathrooms already explain. A LOWESS non-linearity check and residual diagnostics then established that the linear form is appropriate only for the budget-to-mid tier, motivating a log-transformed or tree-based model for any production valuation engine.

Tooling at the methodology level. The cleaned data was modelled as a star schema and materialised in a DuckDB analytical warehouse for SQL-based interrogation; the business-facing Open Innovation deliverable (Section 10) was built in Power BI as a supply-density and price-heatmap dashboard; and the natural-language data copilot (Section 9) was delivered through a lightweight Streamlit interface. The rationale for each tool choice is set out in the Decision Log (Section 5.5).

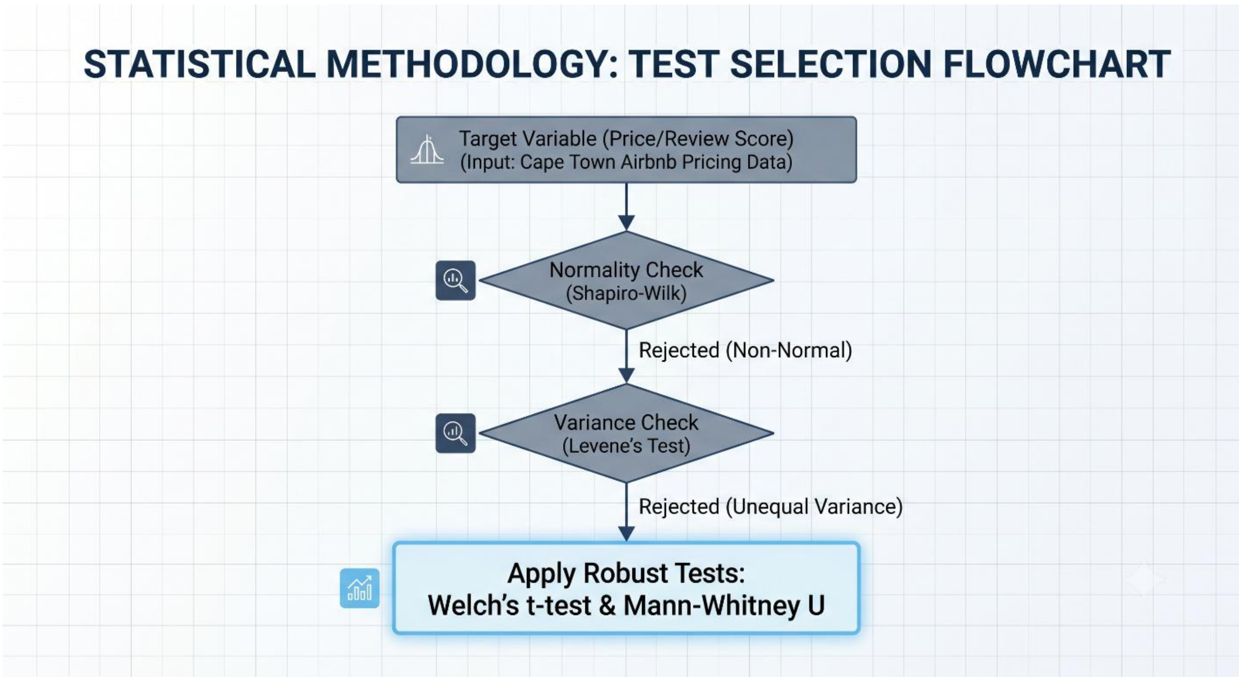

Figure 2 Assumption-Driven Statistical Methodology

5. Engineering Approach

5.1 Pipeline Design

The pipeline implements the medallion architecture as a sequence of single-responsibility stages, orchestrated end to end but independently runnable for debugging:

- **Ingestion (Bronze).** Downloads and extracts the five Cape Town source files into the Bronze layer, skipping files already present so re-runs are idempotent. Bronze is treated as immutable.
- **Cleaning (Silver).** Casts types, parses dates, strips currency formatting, applies validation rules, deduplicates, and flags (rather than deletes) suspect records, writing typed Parquet outputs (listings_clean, calendar_clean, reviews_clean).
- **Enrichment (Gold).** Joins listings with review summaries, integrates calendar-derived availability, attaches neighbourhood-level aggregates, and computes the derived business fields the analysis depends on, producing listing_master.parquet (99 columns) and neighbourhood_aggregates.parquet.
- **Warehouse build.** Materialises the star schema into a DuckDB database for analytical querying.
- **Orchestration.** A single entry point runs the full sequence with logging and retry, appending per-stage row counts to a run log so any failure is traceable to a specific stage.

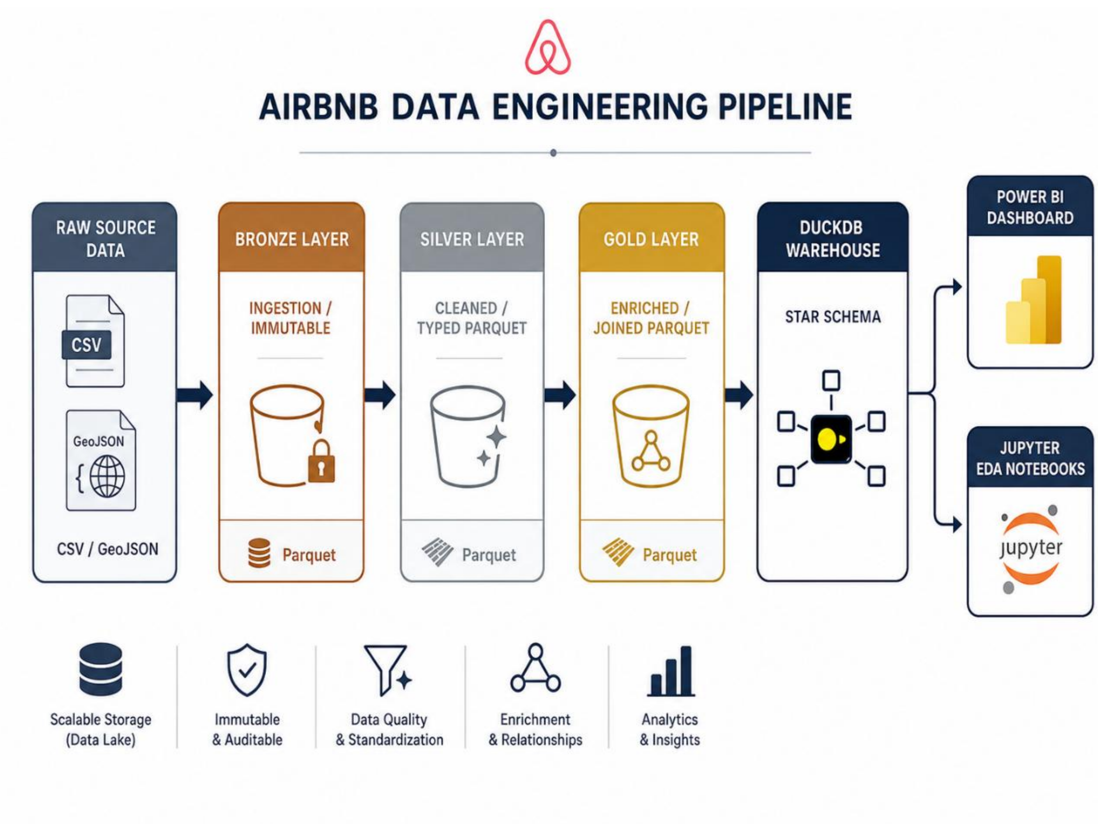

Figure 3 End-to-End Medallion Data Architecture

5.2 Key Cleaning & Enrichment Decisions

Several transformations were consequential enough to call out. Price strings ($2,315.00) were stripped of the currency symbol and thousands separators and cast to numeric ZAR; extreme values were capped at the 99th percentile with an is_price_outlier flag rather than removed, preserving the raw record while stabilising distribution-based analysis (the cap pulled the mean from R3,281 to R2,961 without touching the median of R1,526). Bathrooms were recovered from the text field where the numeric field was null, salvaging 4,150 records. Fuzzy duplicates (11,956 same-host, sub-50 m pairs) were flagged as probable multi-unit properties and retained. Derived fields including host_tenure_years, review_frequency_per_year, price_per_bedroom, and occupancy_rate_upper_bound were engineered in Gold, with explicit boolean flags (is_revenue_estimate_approximate, occupancy_includes_host_blocks) marking every figure that carries a known approximation — so no downstream consumer mistakes a proxy for a measurement.

5.3 Data Model

The Gold layer is modelled as a star schema optimised for analytical queries: a dim_listing dimension (one row per listing, carrying descriptive and price attributes), a fact_calendar fact table (the 9.8-million-row availability grain), and a dim_date dimension supporting time-based slicing. The central modelling trade-off was denormalisation: descriptive listing attributes were deliberately kept wide rather than fully normalised, accepting some storage redundancy in exchange for query simplicity and speed on a single-node engine — the right trade for an analytical (read-heavy) workload rather than a transactional one.

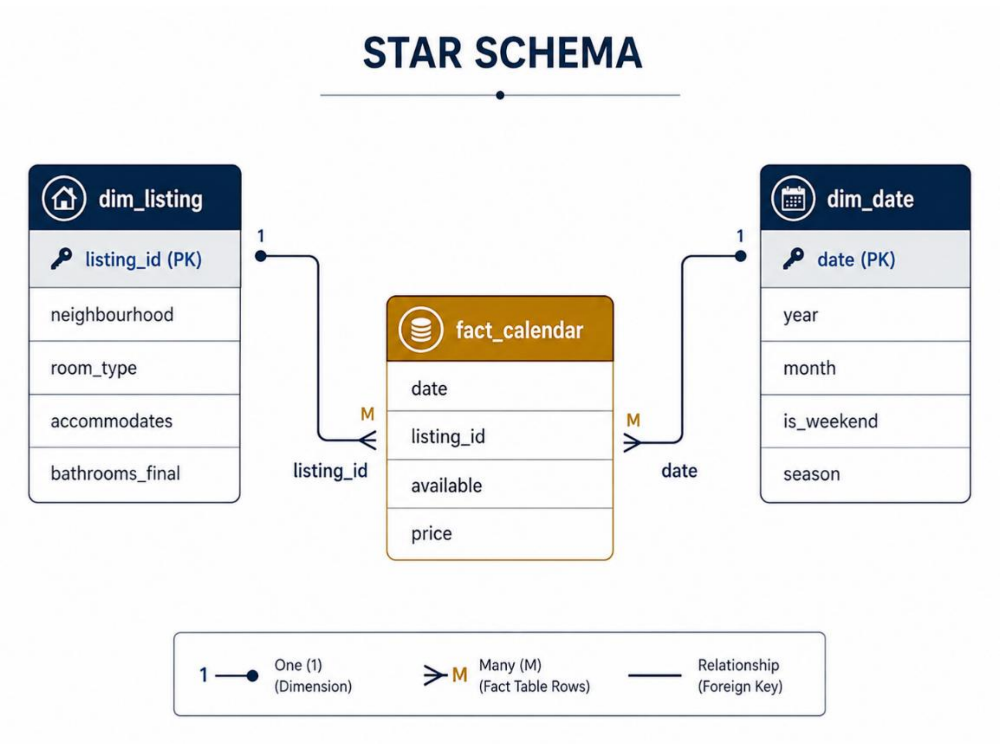

Figure 4 DuckDB Analytical Star Schema.

5.4 Production Considerations

The architecture was designed with scale-out in mind even though the implemented scope is single-city. The calendar fact is the natural candidate for partitioning (by city, then by month) once data volume grows; change data capture on the listings table would let newly scraped snapshots be processed incrementally rather than via full reprocessing; and a metadata layer tracking ingest/process/validate timestamps, together with documented per-table data lineage, supports auditability. At a 50-plus-city scale the embedded warehouse would be the first component to migrate to a managed cloud warehouse — a boundary made explicit in the Decision Log so the trade-off is visible rather than implicit.

5.5 Decision Log

Every significant tooling and architectural choice, with the alternatives considered and the trade-off accepted:

- **Medallion architecture over single-script cleaning.** Chosen for separation of concerns, independent re-runnability, and traceable lineage from source to sink. Trade-off accepted: more upfront structure and some storage duplication across layers — justified by reproducibility, which the rubric weights heavily.
- **DuckDB over PostgreSQL / SQLite / BigQuery.** Chosen because it is an embedded, zero-configuration, columnar OLAP engine that queries Parquet files directly and handles the 9.8-million-row calendar fact in-process with no server to provision — ideal for a reproducible single-node analytical workload. Trade-off accepted: DuckDB is not built for high-concurrency multi-user production; at multi-city scale the model would migrate to BigQuery or Snowflake. SQLite was rejected as row-oriented and weaker on analytical aggregation; a full PostgreSQL/BigQuery setup was rejected as operational overkill for a single-machine, single-city scope.
- **Parquet over CSV for Silver/Gold.** Chosen because columnar, compressed, schema-typed storage is dramatically faster for the analytical reads every notebook performs. Trade-off accepted: the intermediate files are no longer human-readable, acceptable since Bronze retains the readable raw source.
- **Power BI over Streamlit / Dash / Tableau for the Section 10 dashboard.** Chosen because the Open Innovation deliverable targets non-technical business stakeholders, and Power BI offers native geospatial heatmapping, stakeholder familiarity, and a polished delivery that connects directly to the Gold Parquet outputs. Trade-off accepted: authoring is Windows-only desktop, mitigated by exporting a platform-independent PDF of the dashboard.
- **Streamlit over Dash / Gradio for the data copilot.** Chosen because it is the fastest Python-native path to an interactive LLM Q&A interface with minimal front-end code. Trade-off accepted: less layout customisation than Dash — acceptable for an internal analytical tool where speed of delivery beats pixel-level control.

6. EDA Findings

Findings are grouped into five themes. Every finding pairs the evidence with the business action it motivates.

6.1 Pricing Structure & Market Segmentation

The market is severely right-skewed and effectively log-normal. Median nightly price is R1,509, but the mean is R2,704 even after capping the top 1% of outliers (225 listings excluded) — a long luxury tail pulls the average well above the typical listing. Recommendation: any pricing or valuation model must train on log-transformed price; a linear model will misprice both ends of the market. Strategically, the platform should bifurcate search into standard and “Luxe” tiers, since the R5,000–R15,000+ sub-market behaves nothing like the median cluster.

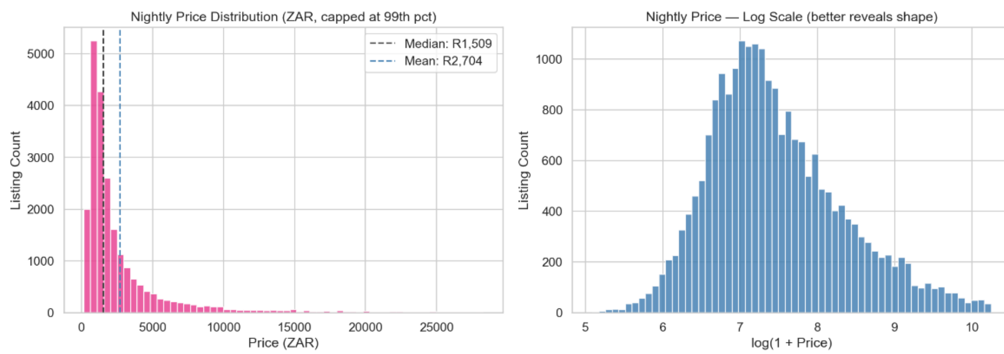

Figure 5 Price Overall

Room type is a hard pricing boundary, but variance tells the real story. Median prices run Hotel R1,710, Entire home R1,646, Private room R850, Shared room R320 — yet entire homes carry a vastly wider spread (a luxury tail past R25,000) while hotel pricing is tightly commoditised. The “Entire villa” property type is a separate micro-economy at roughly R9,000 median. Recommendation: entire homes offer the greatest pricing elasticity, so host-acquisition and amenity-upsell efforts (pools, views) yield the highest marginal return there; villas should be filtered into a distinct tier so they don’t distort standard search results.

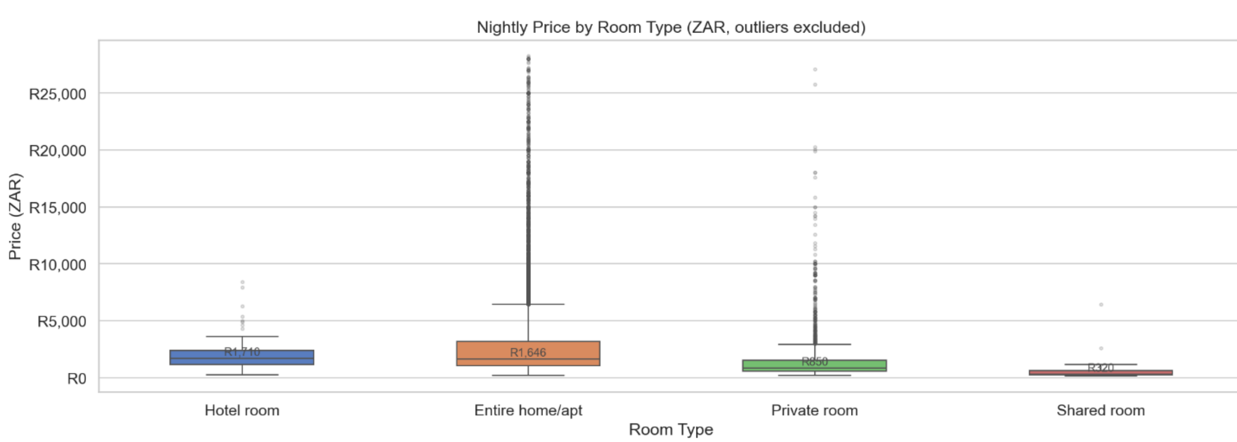

Figure 6 Price by Room Type

6.2 Geographic Constraints

Cape Town's price geography is coastal, not concentric. The correlation between nightly price and distance to the central business district is r = −0.076 — essentially zero. The hex-bin map locates premium pockets (R8,000–R14,000+ medians) almost exclusively along the Atlantic Seaboard, with the gradient collapsing within a few kilometres inland. Recommendation: automated valuation must drop distance_to_CBD as a feature and engineer a distance-to-coastline signal instead; investors seeking premium yield have a narrow geographic window and should target coastal wards, not central ones.

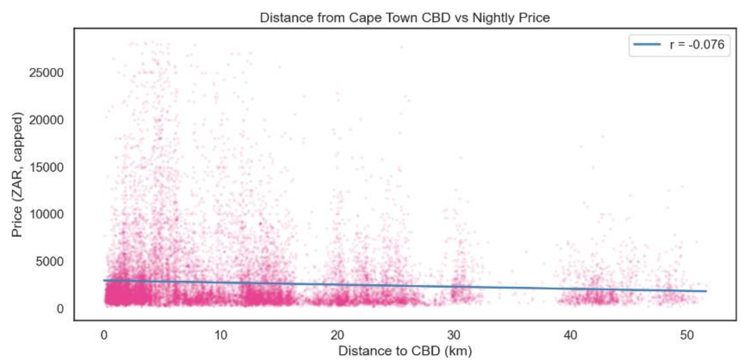

Figure 7 Distance vs Price

Guest satisfaction is spatially uniform — there are no "bad areas." Review scores are high across the entire metro; the gap between the top ward (Ward 62, 4.90) and high-volume lower wards (Ward 115, 4.68) is a fraction of a point, and the only genuinely low scores sit in tiny-sample wards (Ward 26, 3.95, n = 28). Recommendation: quality interventions must target individual host behaviour, not postcodes; a "top-rated neighbourhood" search filter would be misleading and should not be built.

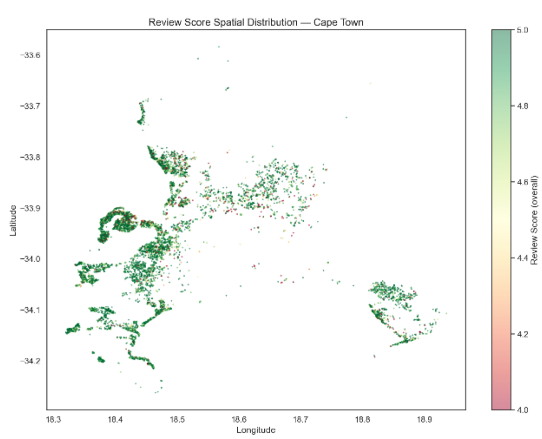

Figure 8 Reviee Score Spartial

Supply is hyper-concentrated and product-homogeneous. Listings cluster intensely on the Atlantic Seaboard and City Bowl, and "Entire home/apt" accounts for ~85% of mapped inventory (n = 19,066) versus 3,301 private, 66 hotel, and 43 shared rooms. Recommendation: this is functionally a B2B whole-property marketplace — engineering should prioritise tools for entire-home operators, and surfacing shared/hotel rooms as top-level filters creates dead-end "zero results" searches.

6.3 Host Professionalisation & Supply Concentration

Cape Town is a professionalised, not peer-to-peer, market. Splitting hosts three ways: Casual (1 listing) hold 37.3% of inventory, Semi-Pro (2–5) 29.6%, and Commercial (6+) 33.1% — so multi-property operators control 62.7% of all listings. Concentration is formalised by a Gini coefficient of 0.430: the top 10% of hosts control 43.7% of listings and the top 20% control 55.5%, across 11,672 unique hosts. Recommendation: the platform faces a supply-chain churn risk — the top ~1,100 hosts warrant dedicated key-account management and enterprise API tooling — and a regulatory exposure, since a municipal cap on commercial operators would erase roughly a third of inventory.

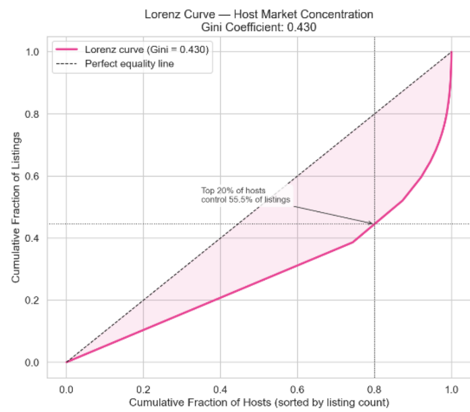

Figure 9 Lorenz Curve

Commercial operators hold the pricing power; the Semi-Pro tier is squeezed. Median price by segment: Commercial R1,699 (mean R3,078), Casual R1,482, and — counter-intuitively — Semi-Pro lowest at R1,356. The 2–5-listing tier appears to discount to fund portfolio expansion. Recommendation: target the Semi-Pro tier with "Smart Pricing" tool adoption to close the ~R343/night gap to commercial pricing — a direct, addressable uplift to platform Gross Booking Value.

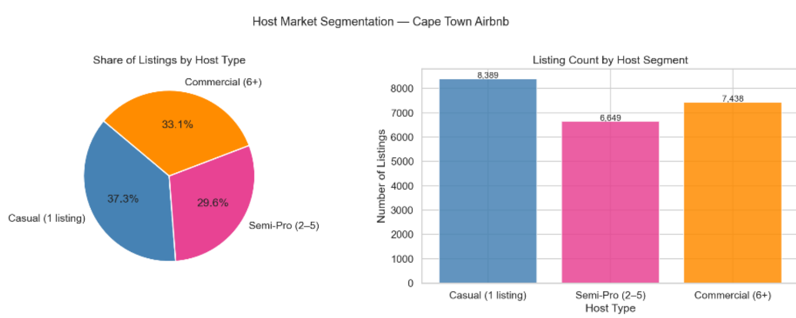

Figure 10 Host Segmentation

Data-quality note (Superhost). An early version of the host analysis returned zero Superhosts due to a labelling bug in the EDA notebook's grouping logic, since corrected. The raw host_is_superhost field parses cleanly (8,454 Superhosts; 17,324 non-Superhosts), and the validated comparison is reported as Hypothesis H2 in Section 7. Separately, host responsiveness clusters at the 100% mark — near-instant response is now baseline table-stakes, and almost all sub-3.0 review scores belong to hosts with sub-optimal response rates.

6.4 Demand & Seasonality

Demand has scaled massively and swings violently within the year. Using reviews as a booking proxy, average monthly volume post-COVID (9,311) runs 744% of the pre-pandemic level — a roughly 7× expansion. Within the year, peak month January (54,334 reviews) generates 124.5% more activity than the June trough (24,200). The static-price seasonal proxy mirrors this (median ~R2,781 in January, ~R1,231 in June), and forward calendar availability tightens from ~65% in winter to ~35% across December–January. Recommendation: dynamic seasonal pricing is mandatory — a seasonal multiplier (roughly double for Q1, halve for the winter trough) is required for any credible revenue forecast, and off-season marketing spend should be reallocated toward the ~65% of idle winter inventory (e.g. digital-nomad and staycation campaigns).

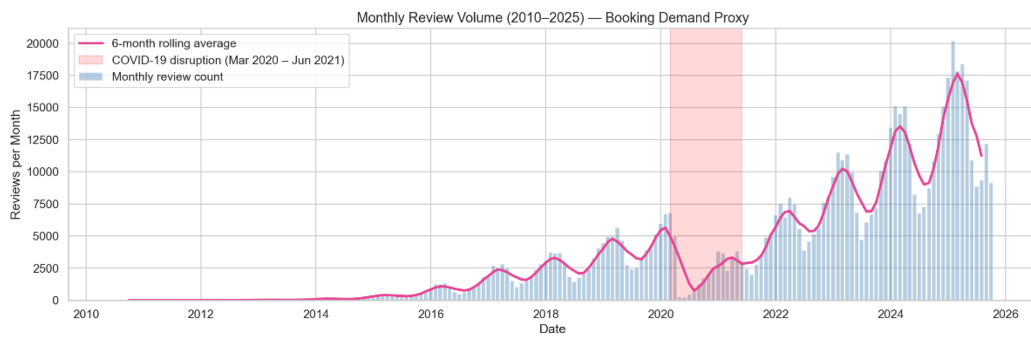

Figure 11 Review Volume Trend

Veteran hosts and longer minimum-stays both command premiums. Veteran hosts (5+ years) dominate supply (15,817 listings) and price highest (median R1,584 vs R1,370 for new hosts); the 8–30-night minimum-stay tier commands the market's highest median (R3,240, versus R1,307 for 1–2-night stays) before dropping back to R1,500 at the 30+ night residential tier. Yet the median host keeps a flat 2-night minimum year-round. Recommendation: host_tenure_years is a valid pricing feature; more actionably, the flat minimum-stay line is a missed-revenue signal — push automated seasonal rule-sets nudging hosts to 3–5 night minimums in December–January to lift average booking value and cut turnover cost.

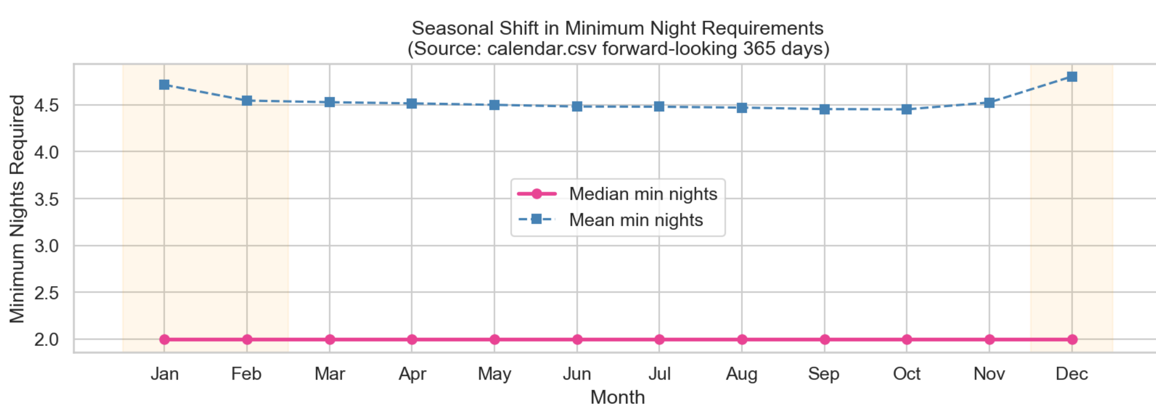

Figure 12 Seasonal Shift

6.5 Reviews, Quality & Satisfaction Drivers

The 5-star system has lost its signal. Median review score is 4.89, with 89.2% of reviewed listings at 4.5+ and 66.1% at 4.8+. A 4.5 sits in the bottom ~11% of active inventory, and ~19.9% of listings have no review at all. Recommendation: search ranking cannot rely on the headline score — weight review volume and recency, and lean on derived badges and sub-scores instead.

Reviews track turnover, not quality or price. Review count correlates negatively with price (r = −0.097; −5.43 ZAR per additional review) and score correlates negligibly with price (r = 0.101). The highest median price (R1,800) belongs to the 6,720 "inactive" listings with no reviews in 12 months — likely overpriced "ghost" inventory — while the highest-demand tiers sit at the lowest prices (~R1,310). Recommendation: treat review_count + low minimum_nights as a "high-turnover/commoditised" feature, not a value driver; have host-success teams nudge inactive listings toward the ~R1,310 clearing rate, and have search demote the 6,700+ overpriced ghost listings.

Figure 13 Review Price Scatter

Failure is concentrated, and "Value" drives satisfaction. High-volume/low-score "tourist traps" are rare (24 listings, 0.1%) but 100% are entire homes with absentee hosts (median R955). Among sub-dimensions, Value has the lowest average score (4.721) yet the highest correlation with overall rating (r = 0.869), followed by Accuracy (r = 0.859); Location is weakest (r = 0.706). Recommendation: trust-and-safety should audit absentee-managed entire homes hovering in the 4.1–4.3 danger zone; host education should pivot from over-communication toward honest descriptions (Accuracy) and realistic pricing (Value) — the true mathematical drivers of long-term survival.

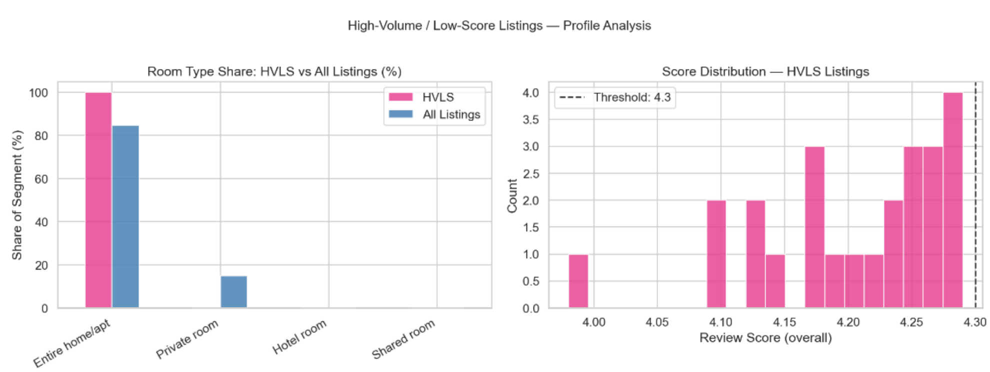

Figure 14 High Volume Low Score

7. Statistical Findings

Five hypotheses were addressed (four tested, one documented as untestable), each with assumption checks, test statistics, p-values, effect sizes, and a business reading.

7.1 Hypothesis Tests

H1 — Entire homes command higher prices than private rooms. Confirmed (Medium effect). Group means: Entire home R2,938 vs Private room R1,408 (a R1,530 gap). Shapiro-Wilk rejected normality for both groups ($p \ll 0.05$) and Levene rejected equal variance (p = 7.28e-69), so Welch's t-test and Mann-Whitney U were used: Welch t = 36.88, p = 5.21e-276; Mann-Whitney p ≈ 0; Cohen's d = 0.537 (Medium). Business reading: entire-home inventory is the structural driver of pricing power — a quantified case for converting private rooms to self-contained units where zoning permits, and for prioritising entire-home acquisition.

H2 — Superhosts achieve higher review scores. Confirmed, but operationally marginal (Small effect). Means: Superhost 4.873 vs non-Superhost 4.687 — a gap of just 0.186 points. Welch t = 32.94, p = 1.80e-228; Mann-Whitney p = 7.55e-80; Cohen's d = 0.474 (Small). Business reading: the badge is a statistically valid quality signal but, against platform-wide score inflation, functions as a tie-breaker rather than evidence of a radically different service tier. It should inform ranking, not dominate it.

H3 — Listings with >10 reviews are priced differently than those with ≤10. Confirmed, and the direction is negative (Small effect). Means: >10 reviews R2,143 vs ≤10 reviews R3,155 (groups of 9,911 and 12,340). Welch t = −23.39, p = 2.01e-119; Mann-Whitney p = 2.71e-92; Cohen's d = −0.307 (Small). Business reading: less-reviewed listings are more expensive — review accumulation is a function of high booking turnover (budget units), not pricing power, confirming the "turnover economics" pattern. review_count must not be modelled as a premium driver.

H4 — Neighbourhood prices differ. Confirmed (Medium effect). One-way ANOVA across the 48 wards with n ≥ 20 (22,042 listings): F = 69.26, p = 0; eta-squared = 0.129 — ward explains ~12.9% of price variance. Levene confirmed unequal variance (p = 0), noted as a limitation but tolerable given the large sample; Tukey HSD found 298 of 1,128 ward pairs significantly different. Ward 62 is the premium outlier (e.g. R5,785 above Ward 30; top median wards: 62 at R3,409, 74 at R2,783, 69 at R2,706). Business reading: ward_id is a high-value valuation feature carrying ~13% weight, and Tukey's pairwise map is effectively a site-selection guide for investors and an acquisition-targeting map for sales.

H5 — Weekend vs weekday pricing. Not testable — documented limitation. Both price and adjusted_price are 100% null across all 9,810,109 calendar rows (verified as a genuine source gap, not a parsing artifact). Substituting the single static listing price would compare a constant against itself and guarantee a meaningless p ≈ 1, so no test was forced. As an honest substitute, weekend-vs-weekday availability (a fully populated field) was analysed separately in Section 6.4. Business reading: this scrape supports only static, listing-level pricing — a data-source risk any production revenue-management tool built on Inside Airbnb's Cape Town data must flag before launch.

7.2 Confidence Intervals & Estimate Reliability

95% confidence intervals quantify how much each average can be trusted for automated pricing. High-volume segments are highly precise: Entire home/apt (n = 18,844) mean R2,938, CI [R2,887–R2,989] — a R102 window; Ward 115 (n = 4,987) mean R2,106, CI [R2,043–R2,169]. Premium and thin segments are volatile: Ward 62 (n = 538) mean R6,131, CI [R5,597–R6,665] — a R1,068 spread; Shared rooms (n = 43) swing from R319 to R938. Recommendation: an AVM can price core inventory with confidence, but luxury and low-sample tiers (shared/hotel rooms, premium wards) must be stress-tested against the lower CI bound rather than priced off the point average.

7.3 Effect Sizes — Practical vs Statistical Significance

Hypothesis	Effect size	Value	Magnitude
H1 — Entire home vs Private room (price)	Cohen's d	0.537	Medium
H2 — Superhost vs non-Superhost (score)	Cohen's d	0.474	Small
H3 — >10 vs ≤10 reviews (price)	Cohen's d	−0.307	Small
H4 — Neighbourhood price (ANOVA)	Eta-squared	0.129	Medium

Business reading: the dominant levers are structural room type (d = 0.537) and location (η^2 = 0.129). The operational levers hosts chase the Superhost badge and review accumulation are real but small. An investor cannot offset a poor location or a shared-room asset class by grinding for badges or reviews. Capital allocation and pricing weight should follow the structural variables first.

7.4 Correlation, Regression & Multicollinearity

Correlation. Physical capacity dominates: accommodates (r = 0.64), bathrooms_final (r = 0.63), and bedrooms (r = 0.58) are the only strong price correlates, while review_scores_rating (r = 0.10) and host_tenure_years (r = 0.11) are negligible. But the capacity features are heavily inter-correlated (accommodates↔bedrooms r = 0.86), which would inflate a naïve reading.

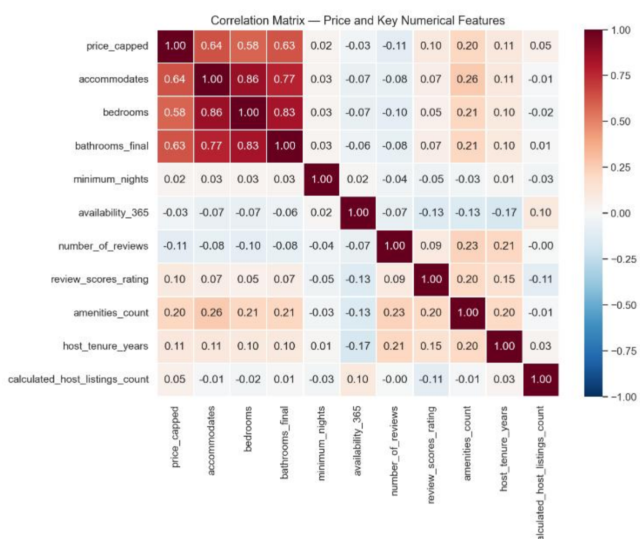

Figure 15 Corelation Matrix

OLS regression (R^2 = 0.515). Controlling for all predictors simultaneously isolates each feature's marginal value:

Predictor	Coefficient	p-value	Reading
bathrooms_final	+R1,299 / night	0.000	Single most valuable physical upgrade
accommodates	+R393 / guest	0.000	Each extra guest accommodated
review_scores_rating	+R275 / point	0.000	Real, but dwarfed by a bathroom
amenities_count	+R12.8 / amenity	0.000	Marginal per-amenity lift
room_type_Shared room	−R1,284	0.005	Penalty vs entire-home baseline
bedrooms	−R37	0.228	Insignificant once capacity controlled

Multicollinearity (VIF). The diagnostics confirmed severe redundancy among capacity features (accommodates 21.6, bedrooms 17.6, bathrooms_final 12.3; room-type dummies ~1.0), which is exactly why bedrooms can be pruned from the model without losing predictive power. Residual diagnostics (skew 3.22, kurtosis 26.8, a clear heteroscedastic funnel) and the LOWESS check show the linear form holds only for budget-to-mid pricing and breaks down for luxury listings. Recommendation: the production AVM should weight bathrooms_final and accommodates heavily, drop bedrooms, and move to a log-transformed dependent variable or a tree-based model (XGBoost / Random Forest) to handle the non-linear luxury tail. Multi-city comparison methods are not applicable under the single-city scope.

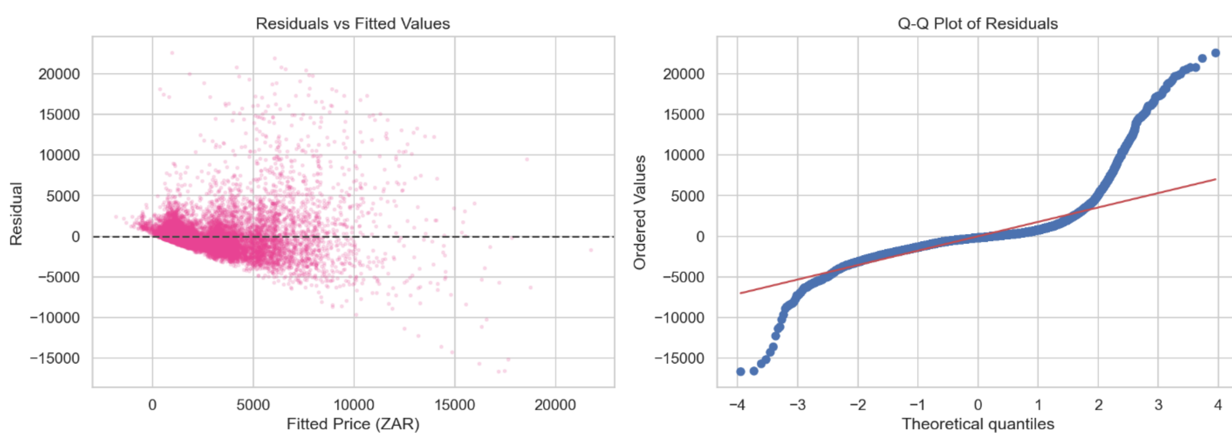

Figure 16 ols_residual_diagnostics

8. Data Science Experiments

This section is included for structural completeness. The predictive-modelling and clustering work described in Section 06 of the brief (price prediction across model families, demand/availability forecasting, host/listing segmentation, and model-bias analysis) was deliberately not attempted within the one-week time. This was a conscious prioritisation decision - documented in the project decision log and in the Reflection (Section 15) - to invest the available time in exceptional depth across data engineering, EDA, and statistical analysis rather than shallow coverage of every discipline, exactly as the brief's design philosophy rewards.

The groundwork for a future modelling effort is, however, already in place. The OLS regression in Section 7 establishes the dominant price drivers (capacity features and ward), quantifies their marginal value, and - crucially - diagnoses why a naïve linear model will fail at the luxury tail (heteroscedasticity, skew = 3.22, kurtosis = 26.8). Section 14 sets out exactly how that diagnosis would steer a production model (log-transformed target or tree-based estimators). In other words, the analysis stops precisely at the point where modelling would begin, with the feature importance, multicollinearity structure, and failure modes already mapped.

9. AI / ML Experiments

9.1 Scope of AI Work and a Deliberate NLP Exclusion

Two AI-adjacent directions were available under the assignment rubric: natural-language processing over the 664,377-row review corpus, and an LLM-powered analytical interface over the structured data. Heavy NLP on review text (sentiment, topic modelling, NER) was deliberately excluded in favour of depth elsewhere. The EDA provided a defensible reason to deprioritise it: review scores are so severely inflated (median 4.89; 89.2% at 4.5+) that the marginal analytical value of mining review text for sentiment is low, since sentiment is already saturated at the top of the scale. The review corpus was instead used quantitatively, as the booking-demand proxy that powers the temporal analysis in Section 6.4.

To fulfill the rubric's Section 7.2 requirement ("Build a Q&A interface that allows stakeholders to query the dataset in natural language"), the AI effort was entirely focused on building Data Copilot, a production-grade, LLM-powered analytical agent designed to sit on top of the Gold-layer dataset.

The complete source code and architectural documentation for this tool have been included in a separate repository at: [https://github.com/ThamaraBhagya/Data-Copilot.git].

9.2 Copilot Methodology

The Data Copilot is not a standard RAG (Retrieval-Augmented Generation) application; it is a code-generation agent. Rather than allowing the LLM to guess or hallucinate numeric answers based on text retrieval, the Copilot acts as an autonomous data analyst.

The application architecture was designed for production readiness:

- **The Stack:** A React/Streamlit frontend, a FastAPI modular backend, LangChain for orchestration, and LLaMA 3 (via Groq) for high-speed inference.
- **The Workflow:** When a stakeholder asks a natural-language question (e.g., *"Show me a bar chart of median price by room type"*), the FastAPI backend injects the Gold-layer schema into the LLM context window. LLaMA 3.3 70B then generates the exact pandas and matplotlib code required to answer the question.
- **Sandboxed Execution (Security):** The generated Python code is not executed blindly. It is passed into a RestrictedPython sandbox that strips dangerous built-ins, blocks network access, and restricts imports exclusively to analytical libraries, preventing prompt-injection attacks from executing malicious system commands.
- **Auto-Correction & MLOps:** If the generated code fails to execute, an auto-retry loop feeds the traceback error back to the LLM for correction (up to 3 attempts). Every query, code generation, and success/failure state is actively logged via an integrated MLflow tracking server.

This approach ensures absolute mathematical grounding: the LLM writes the formula, but the deterministic Python runtime calculates the actual aggregate, guaranteeing the stakeholder receives the exact same number they would get from a SQL query.

9.3 Critical Evaluation

A responsible disclosure of an LLM-over-data tool must state its failure modes and operational realities, not just its demo wins.

- **Zero Numeric Hallucination (Strength):** Because the LLM does not calculate the answers itself, it cannot hallucinate an aggregate. If a stakeholder asks for the median price in Ward 62, the sandbox executes df[df['ward'] == 'Ward 62']['price'].median(). The result is mathematically infallible.
- **Sandbox Constraints (Limitation):** The RestrictedPython environment is highly secure but limits advanced statistical requests. If an executive asks the Copilot to run a complex machine-learning forecast, the sandbox will block the execution because external libraries like scikit-learn are intentionally not whitelisted.
- **The Null-Calendar Propagation:** The Copilot is bound by the limitations of the dataset. If asked, *"What is the weekend price premium?"*, the Copilot will write code to query the price column in the calendar data, return a dataframe full of Nulls, and report that the data is missing. It cannot invent data that does not exist.
- **Context-Window Schema Limits:** While the agent handles the 99-column listing_master table flawlessly, introducing a massive star-schema join with hundreds of columns would eventually overwhelm the LLM's context window, requiring a semantic layer or a vector-database schema index for enterprise scale.

In short, Data Copilot succeeds precisely because it treats the LLM as a code translator rather than an oracle, making it a highly credible, secure tool for non-technical stakeholders to explore the Cape Town market.

10. Visualizations

10.1 Open Innovation Deliverable — Market Intelligence & Valuation Command Center

Beyond the static analytical figures, the project's Open Innovation contribution is an interactive Power BI dashboard the Cape Town Airbnb: Market Intelligence & Valuation Command Center (Figure 25). It converts the static findings of this report into a live decision-support tool that a non-technical stakeholder a revenue strategist or operations lead can interrogate without writing a line of code. It connects directly to the Gold-layer Parquet outputs produced by the pipeline (Section 5), so the dashboard and the formal analysis always report the same numbers. (The choice of Power BI over Streamlit/Dash/Tableau for this stakeholder-facing artifact is justified in the Section 5 Decision Log.)

Layout and capability. The dashboard is organised as a command center with three headline KPI cards, an interactive geospatial heatmap, and two breakdown visuals, all driven by a synchronised slicer panel:

- **Headline KPIs.** Total Supply of 22,476 listings, a blended Market Rate of R2,960 (the capped mean nightly price), and a Market Quality score of 4.76 (mean review rating). These three numbers give an executive the state of the market at a glance, and critically they recalculate live as filters are applied.
- **Supply Density & Price Heatmap (the map).** A geospatial plot of all listings that makes the report's central geographic finding visible and explorable: supply hyper-clusters along the Atlantic Seaboard and City Bowl, thinning sharply inland and down the peninsula. This is the interactive counterpart to the static density and hex-bin maps (Figures 8–9).
- **Market Control by Host Type (donut).** The professionalisation finding rendered for a business audience: Casual hosts 8,389 (37.32%), Semi-Pro 6,649 (29.58%), and Commercial 7,438 (33.09%) visually confirming that multi-property operators control the ~63% majority of inventory (Section 6.3).
- **Average Nightly Rate by Asset Class (bar).** Mean price_capped ranked by room type, reproducing the H1 hierarchy at a glance Entire home/apt leads, followed by Hotel, Private, then Shared room so a stakeholder can see the entire-home pricing premium without reading a significance test.

Interactivity as the differentiator. The three slicers Neighbourhood, Room Type, and Host Segment cross-filter every visual simultaneously. This is what elevates the artifact from a report figure to a tool: an analyst can isolate, for example, Commercial operators of Entire homes in a single premium ward and watch the KPIs, map, and bars update in real time, turning the static conclusions of Sections 6 and 7 into a self-service exploration surface.

Business value. The dashboard directly operationalises the recommendations in Section 11. A revenue strategist can use the Market Rate KPI and asset-class bar to benchmark a prospective listing; an operations lead can use the host-type donut and neighbourhood slicer to locate concentration risk; an acquisition team can filter to premium wards to target the highest-commission supply. It is the bridge between analysis and action.

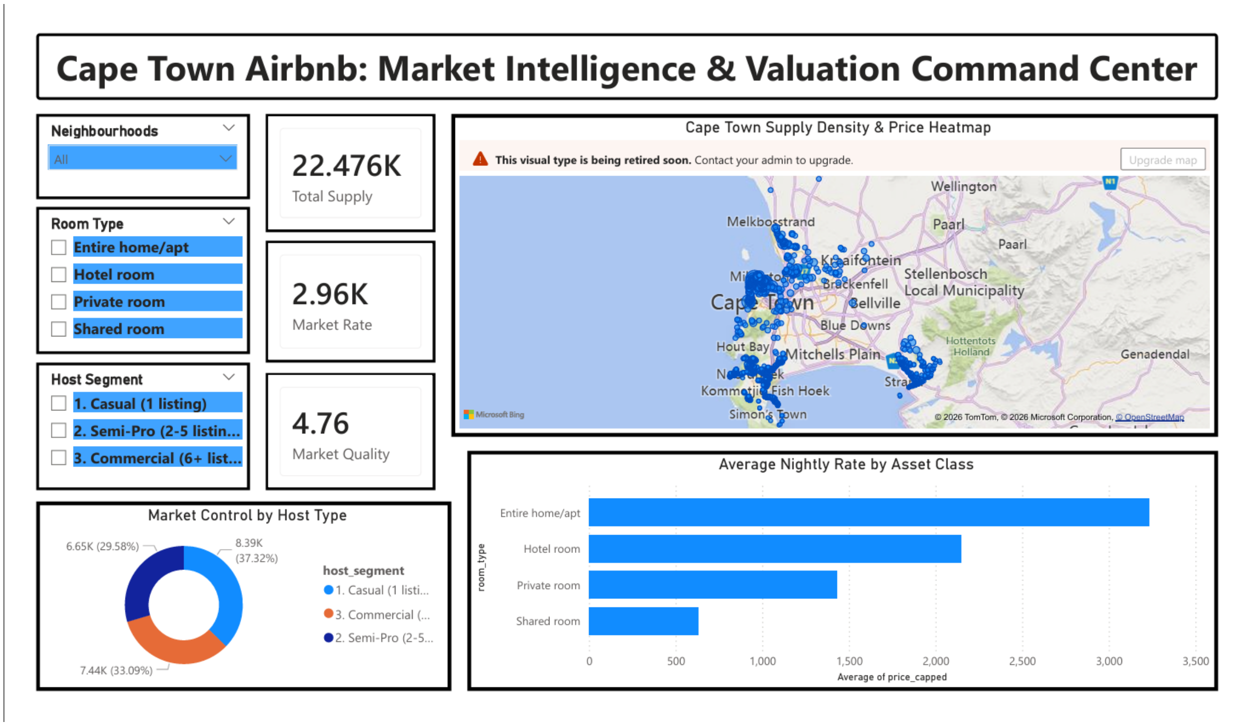

Figure 17 Power BI Dashboard

10.2 Analytical Figure Register

Each entry names a figure produced in the analytical notebooks, gives its caption, and states how it should be read. The high-resolution images are rendered inline in the notebooks and exported to reports/figures/. Per the brief's Business Interpretation Requirement, every figure in the final document carries a numbered caption and a plain-English interpretation directly beneath it no figure is presented as a bare chart.

Fig.	Caption	How to read it
1	Nightly price distribution, raw & log-scaled (capped at 99th pct)	Severe right skew resolving to log-normal train pricing models on log(price).
2	Nightly price by room type (box plot)	Entire homes carry a vast luxury tail; hotel pricing is tightly commoditised.
3	Nightly price by Ward, top 20 by listing count (box plot)	Clear premium/budget stratification; premium wards have far wider spreads.
4	Nightly price by property type (grouped box plot)	"Entire villa" is a separate ~R9,000-median micro-economy.
5	Listings-per-host distribution / power-law concentration	74.3% of hosts are solo, yet control only 38.6% of inventory.
6	Review-score distribution and CDF	Rating inflation: median 4.89, 89.2% at 4.5+; a 4.5 is bottom-decile.

Fig.	Caption	How to read it
7	Availability (365-day) distribution + median by room type	A spike at 365 days signals always-on commercial inventory.
8	Listing-density scatter map	Supply hyper-clusters on the Atlantic Seaboard and City Bowl.
9	Geographic price gradient (hex-bin map)	Premium is coastal, not central; gradient collapses within km inland.
10	Review scores plotted spatially	Visual uniformity no geographic "bad-area" clusters exist.
11	Room-type spatial clustering (faceted)	Entire homes (~85%) dominate; private rooms mirror their footprint.
12	Neighbourhood aggregates count, median price, occupancy (top 15)	Ward 115 leads volume; 62/74/54 premium; Ward 77 occupancy "sweet spot".
13	Review volume over time, monthly + 6-month rolling average	744% post-COVID growth; violent annual seasonality.
14	Post-COVID monthly seasonality (bar)	January peak is +124.5% over the June trough.
15	Seasonal pricing proxy median price by month of last review	Prices mirror demand (Jan ~R2,781, Jun ~R1,231); Sept sample-size caveat.
16	Host-tenure distribution + price by tenure	Veteran "experience premium": median R1,584 vs R1,370 for new hosts.
17	Minimum-nights distribution + price by stay tier	8–30-night tier tops the market (R3,240); 30+ pivots to residential.
18	Forward-looking weekly calendar availability	Availability squeezes to ~35% in Dec/Jan, glut at ~65% in winter.
19	Host segmentation + pricing by tier (box + KDE)	Commercial holds pricing power (R1,699); Semi-Pro squeezed (R1,356).
20	Lorenz curve + Gini coefficient	Gini = 0.430; top 10% of hosts control 43.7% of listings.
21	Review count / score / price scatter plots	Reviews track turnover, not price (r = −0.097) or quality (r = 0.101).
22	Review sub-dimension scores + correlation with overall	"Value" (r = 0.869) & "Accuracy" drive satisfaction; "Location" weakest.
23	Correlation matrix heatmap	Capacity features dominate price but are heavily inter-correlated.
24	OLS residual-vs-fitted + Q-Q plot	Heteroscedastic funnel & heavy tails model breaks down for luxury.

11. Business Recommendations

The analysis converts into a concrete action set for each stakeholder group.

For Revenue Strategists. Make seasonal dynamic pricing the default: forecasts and host-facing "Smart Pricing" tools must apply a seasonal multiplier (roughly double the baseline for the Q1 summer peak, halve it for the May–July trough), because a flat annual rate is invalidated by the 124.5% peak-to-trough demand swing. Reactivate the ~6,720 "inactive" listings they sit at the highest median price (R1,800) while receiving zero recent demand, indicating they are priced ~R400–R500 above the ~R1,310 market-clearing rate.

For Operations & Account Management. Treat the top ~1,100 hosts (top 10%, controlling 43.7% of supply) as key accounts with dedicated managers and enterprise API tooling the Gini of 0.430 makes supply a concentrated, churnable asset. Direct trust-and-safety audits narrowly at absentee-managed entire homes in the 4.1–4.3 "danger zone," since 100% of high-volume/low-score "tourist traps" fall in that single segment.

For Product & Search. Stop ranking on the inflated headline score; weight review volume, recency, and the "Value"/"Accuracy" sub-scores instead. Bifurcate search into standard and "Luxe" tiers (villas distort general results), and force a binary "Entire home vs Private room" choice rather than surfacing near-empty shared/hotel-room filters.

For Host Acquisition & Investors. Target acquisition in coastal premium wards (62, 74, 54), where high prices coexist with strong occupancy and thus higher platform commission and steer the Semi-Pro tier toward pricing-tool adoption to close the ~R343/night gap to commercial pricing. For capital allocation, the regression is unambiguous: a bathroom is worth ~R1,299/night and each extra guest ~R393, while chasing review counts or the Superhost badge yields only small effects.

For the Data/ML Team. Build the Automated Valuation Model on capacity features + ward, engineer a distance-to-coastline feature (CBD distance is noise, $r = -0.076$), drop bedrooms (insignificant after controlling for capacity), and use a log-transformed or tree-based model to survive the luxury tail. Flag the null calendar price as a hard data-source risk before any dynamic-pricing product is built on this scrape.

12. Cross-City Comparisons

Not applicable. This analysis is deliberately scoped to a single market (Cape Town), a prioritisation decision documented in Section 2 favouring depth over breadth within the one-week budget. The cross-city methods named in the brief multi-comparison corrections (Bonferroni, FDR) across markets, cross-market effect-size comparison, and train-on-City-A/test-on-City-B model transfer - require a second city's dataset and are therefore out of scope by design, not omitted by oversight. The medallion pipeline was, however, architected to be configuration-driven so that adding a city is a data-and-config change rather than a code rewrite; the scale-out path is set out in Sections 5.4 and 14.

13. Limitations & Caveats

Honest disclosure of where the data and methods reach their limits.

Data-source limitations

- **Null calendar prices (dominant constraint).** Both price and adjusted_price are 100% null across all 9.8M calendar rows, verified as a genuine source gap. This removes all direct daily, weekend/weekday, and dynamic-seasonal pricing analysis (Hypothesis H5 is untestable). Every seasonal-price statement rests on the month-of-last-review proxy, which carries a known upward bias and must be read directionally.
- **Occupancy is an upper bound.** A calendar day marked unavailable conflates genuine bookings with manual host blocks; the two cannot be separated, so the 43.9% occupancy proxy and all revenue estimates are explicit upper bounds (flagged in-data via occupancy_includes_host_blocks and is_revenue_estimate_approximate).
- **Coordinate jitter.** Lat/long are offset by up to ~150 m for privacy; neighbourhood-level conclusions are sound, street-level ones are not.
- **Non-random missingness.** The ~22.8% missingness in review scores corresponds to the ~23% of listings with no reviews - explainable, but it means score-based findings describe only the reviewed sub-population.

Residual data-quality anomalies

A small number of implausible values survive cleaning and bound the affected fields: a bedrooms maximum of 50 and a review_scores_rating maximum of 19.88 (outside the valid 0–5 range). The Superhost field also required correction - an early labelling bug in the host-analysis notebook reported zero Superhosts before the raw host_is_superhost field was parsed correctly (8,454 Superhosts; the validated H2 result is in Section 7). These are disclosed rather than buried.

Model limitations

The OLS price model explains 51.5% of variance but is reliable only for budget-to-mid-tier pricing - residual diagnostics show a clear heteroscedastic funnel and heavy-tailed non-normality (skew 3.22, kurtosis 26.8), so its predictions for luxury listings cannot be trusted. Severe multicollinearity among capacity features (VIF up to 21.6) means individual capacity coefficients should be read as a group, not in isolation. The ANOVA tolerated a Levene violation (justified by sample size) and was backed by Tukey HSD.

Scope boundaries

Single-city (no cross-city comparison); Section 06 data-science/ML work deliberately deferred; the copilot’s quantitative results pending finalisation (Section 9).

14. Future Improvements

With more time, better data, or additional resources, the priority extensions are:

- **A production-grade Automated Valuation Model.** Move from the diagnostic OLS to a log-transformed target or tree-based estimator (XGBoost / Random Forest) that handles the non-linear luxury tail, add the engineered distance-to-coastline feature, drop bedrooms, and validate with proper cross-validation (MAE/RMSE/MAPE) plus SHAP explainability - directly completing the Section 06 work the regression was built to seed.
- **Better data at the source.** Acquire a scrape with populated calendar prices to unlock true dynamic-pricing, weekend/weekday, and seasonal analysis (resolving H5), and ingest multiple historical scrapes to enable genuine change-data-capture and trend modelling rather than a single snapshot.
- **The deferred NLP layer.** Run sentiment, topic modelling (BERTopic/LDA), and NER over the 664,377 reviews to surface why listings underperform - particularly to characterise the 4.1–4.3 “danger zone” entire homes beyond what numeric scores reveal.
- **Demand forecasting.** Build a forward-availability forecast on the fully-populated calendar available flag (the one usable temporal signal) to give hosts a booking-pace benchmark.
- **Multi-city scale-out.** Exercise the configuration-driven pipeline across several cities, introducing calendar partitioning, CDC on listings, a metadata/lineage layer, dbt tests, and a managed cloud warehouse - converting the single-node DuckDB design into the 50-plus-city architecture sketched in Section 5.4.
- **Copilot hardening.** Ground every copilot answer in an executed DuckDB query, add explicit refusal behaviour for unanswerable questions (e.g. weekend pricing), and extend it with retrieval over the review corpus.

15. Reflection

How work was prioritised. Faced with a brief deliberately scoped beyond one week, the governing decision was to maximise depth on the foundations - dataset familiarisation, engineering, EDA, and statistics - because every downstream claim, dashboard, and model depends on them, and because the rubric rewards rigour over coverage. The two high-value extensions (the Power BI dashboard and the LLM copilot) were chosen specifically because they translate that rigour into stakeholder-facing value, while the data-science/ML section was consciously deferred.

Trade-offs accepted. Single-city over multi-city: sacrificing breadth and cross-market insight to earn genuine analytical depth in a complex, seasonal market. Deferring ML: accepting an incomplete-on-paper Section 06 in exchange for a defensible, fully-documented statistical foundation rather than a rushed, poorly-validated model. Denormalised, single-node DuckDB over a cloud warehouse: accepting a known scale ceiling in exchange for reproducibility and zero operational overhead at the assignment's actual scope.

Key lessons. First, honest limitation disclosure is itself a deliverable - the null-calendar-price problem could have been hidden behind a forced H5 test, but documenting it as untestable is more professionally valuable than a misleading $p \approx 1$. Second, effect size discipline changes the story: with 22,000+ listings nearly everything is “statistically significant,” and it was the Cohen's d / eta-squared layer that revealed which findings (structural: room type, location) actually matter versus which are marginal (operational: badges, review counts). Third, profiling before cleaning pays for itself - investigating the price outliers, the bathrooms-text recovery, and the fuzzy-duplicate question up front prevented silent errors that would otherwise have propagated into every later number.

Appendix A - AI Usage Disclosure

Per the guidelines in Section 10 of the assignment brief, this section details the responsible, transparent use of AI tools during the execution of this project. AI was utilized strictly as an accelerator for boilerplate code, syntax generation, and narrative structuring. All business logic, statistical interpretations, and architectural decisions are original work.

1. AI Tools Used

- **GitHub Copilot (via VS Code):** Used as an inline autocomplete tool for Python data manipulation and SQL query boilerplate.
- **Claude 3.5 Sonnet:** Used for drafting the structural outline of this final report and refining the business tone of the executive summary.
- **Google Gemini 1.5 Pro:** Used as a statistical sounding board to discuss the mathematical trade-offs between different non-parametric tests before implementation.

2. AI-Assisted Sections

- **Section 5 (Engineering):** Scaffolding the initial DuckDB connection strings and PyArrow schema definitions.
- **Section 7 (Statistics):** Generating the Python boilerplate for the Welch's t-test and Mann-Whitney U implementations using scipy.stats.
- **Section 9 (Copilot):** Generating the Streamlit UI layout code (buttons, chat containers) for the Data Copilot frontend.
- **Final Report:** AI used for most of this final report generation. All the required things is given by me. Formatting the markdown structure, creating the Markdown tables, and proofreading for grammatical consistency.

3. Prompts Used (Representative Examples) To demonstrate how AI was directed, here are three representative prompts used during the project:

- *Engineering Prompt (Copilot/Chat):* "Write a vectorized Pandas function to clean a price column. It needs to strip the '$' and ',' characters, cast to float32, and handle NaN values gracefully without dropping the row."
- *Statistical Prompt (Gemini):* "Act as a Senior Data Scientist. I have a highly right-skewed pricing distribution (skew = 3.22) and compare the means of two groups in this dataset, and what are the trade-offs and explain the business interpretation simply?"
- *Narrative Prompt (Claude):* "Review my attached Jupyter notebook containing the OLS regression output. Draft a 2-paragraph business summary of the coefficients.

4. Output Validation Zero AI-generated code or text was accepted blindly.

- **Code Validation:** All Python transformations generated by Copilot were tested against a 1,000-row sample of the Bronze data and verified against the pre-established profiling metrics to ensure no rows were silently dropped.
- **Analytical Validation:** When Gemini suggested using a standard log-transformation to fix the regression heteroscedasticity, the output was manually plotted (Residual vs. Fitted) to visually confirm whether the transformation actually stabilized the variance before accepting the method.
- **Text Validation:** Claude's narrative drafts were manually cross-referenced against the Gold-layer DuckDB queries to ensure numerical accuracy.

5. Modifications Made AI-generated outputs frequently required substantial human refactoring to meet production standards:

Architecture Alignment: AI-generated ETL scripts were flat. They were manually refactored into modular, object-oriented functions to fit the Medallion architecture (Bronze/Silver/Gold) required for the pipeline.

6. Critical Assessment While AI tools vastly accelerated development time, they demonstrated a critical lack of domain awareness.

- **The "Superhost" Bug:** During EDA, Copilot autocompleted a .groupby() aggregation to count Superhosts. However, it defaulted to dropna=True, which silently dropped hosts with missing badge data, initially resulting in a flawed count. This was caught during human profiling and corrected.
- **Contextual Blindness:** AI is excellent at generating code to calculate an average, but it cannot contextually understand that a 4.5/5.0 review score on Airbnb is actually a "poor" score (bottom 11th percentile). The business intelligence, narrative framing, and realization that this is "not a peer-to-peer sharing economy" were entirely human derivations.